Atividade Avaliativa - Denavit-Hartemberg e RoKiSim

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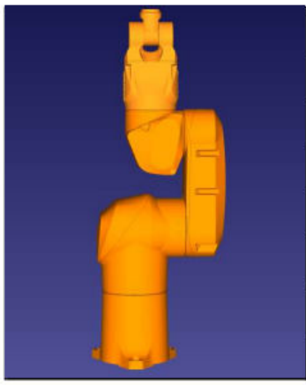
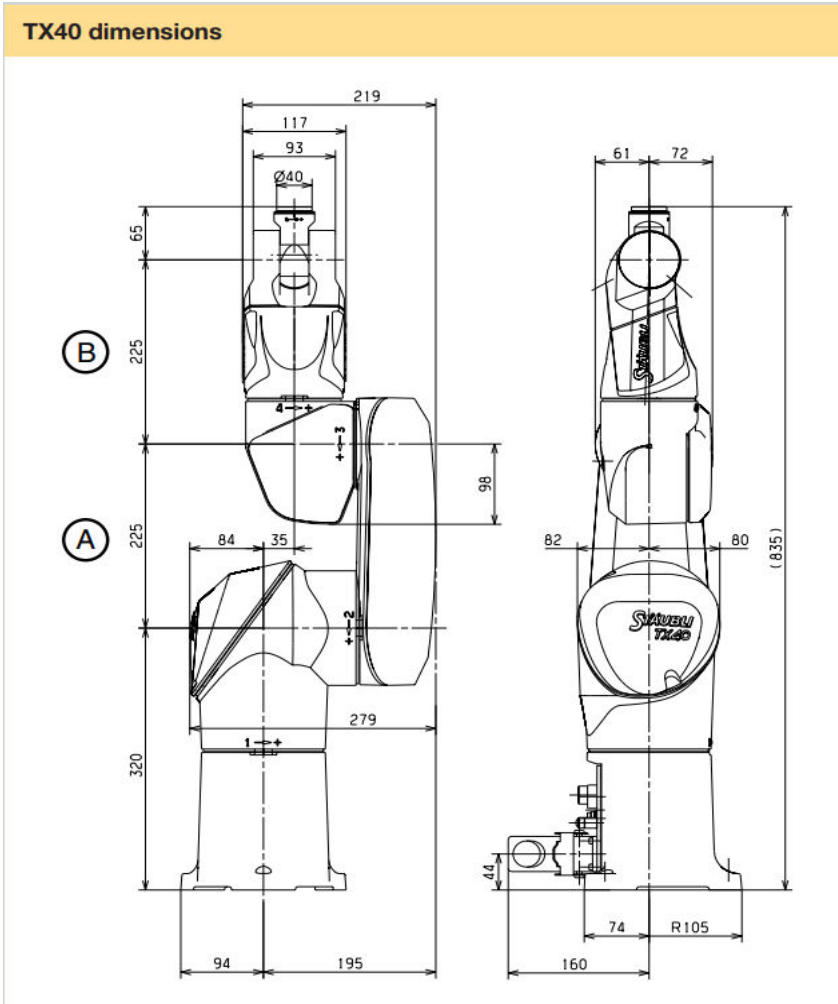
```
% Limpeza da Área de Trabalho
clc; close all; clear all;
```

Manipulador Staubli TX40



Staubli TX40				
Link	**a**	**alfa**	**d**	**teta**
0 - 1	0	-90	320	teta1
1 - 2	225	0	35	teta2 - 90
2 - 3	0	-90	0	teta3 - 90
3 - 4	0	90	225	teta4
4 - 5	0	-90	0	teta5
5 - 6	0	0	65	teta6 + 180




```
% Validação 1 (angulos iguais a zero):
teta1 = 0;
teta2 = 0;
teta3 = 0;
teta4 = 0;
teta5 = 0;
teta6 = 0;
fprintf('Staubli TX40 - Validação 1 \n');
```

```
Staubli TX40 - Validação 1
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 0 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = 0 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais TX40 (Validação 1):')
```

```
Matrizes parciais TX40 (Validação 1):
```

```
% Matriz parcial 0-1:
TX40_1_dh_01 = denavit(0, -90, 320, teta1)
```

```
TX40_1_dh_01 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0   320
     0     0     0     1
```

```

% Matriz parcial 1-2:
TX40_1_dh_12 = denavit(225, 0, 35, teta2-90)
```

```
TX40_1_dh_12 = 4×4
     0     1     0     0
    -1     0     0  -225
     0     0     1    35
     0     0     0     1
```

```

% Matriz parcial 2-3:
TX40_1_dh_23 = denavit(0, -90, 0, teta3-90)
```

```
TX40_1_dh_23 = 4×4
     0     0     1     0
    -1     0     0     0
     0    -1     0     0
     0     0     0     1
```

```

% Matriz parcial 3-4:
TX40_1_dh_34 = denavit(0, 90, 225, teta4)
```

```
TX40_1_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   225
     0     0     0     1
```

```

% Matriz parcial 4-5:
TX40_1_dh_45 = denavit(0, -90, 0, teta5)
```

```
TX40_1_dh_45 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
TX40_1_dh_56 = denavit(0, 0, 65, teta6+180)
```

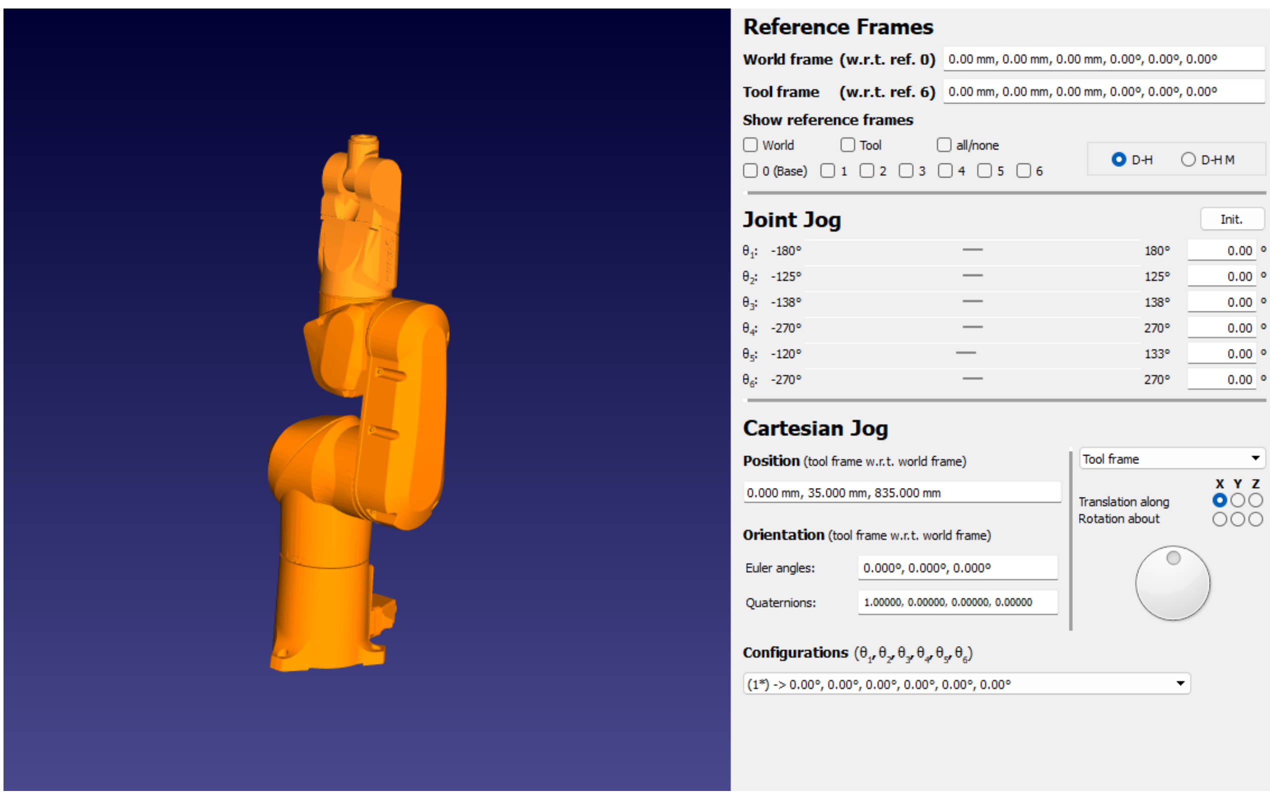
```
TX40_1_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1    65
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante Staubli TX40 (Validação 1):')
```

```
Matriz Resultante Staubli TX40 (Validação 1):
```

```
TX40_1_MR = TX40_1_dh_01 * TX40_1_dh_12 * TX40_1_dh_23 * TX40_1_dh_34 *
TX40_1_dh_45 * TX40_1_dh_56
```

```
TX40_1_MR = 4×4
     1     0     0     0
     0     1     0    35
     0     0     1   835
     0     0     0     1
```




```
% Validação 2 (angulos teta3 = 45 e teta5 = -90):
```

```
teta1 = 0;
teta2 = 0;
teta3 = 45;
teta4 = 0;
teta5 = -90;
teta6 = 0;
fprintf('Staubli TX40 - Validação 2 \n');
```

```
Staubli TX40 - Validação 2
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 45 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = -90 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais TX40 (Validação 2):')
```

```
Matrizes parciais TX40 (Validação 2):
```

```
% Matriz parcial 0-1:
TX40_2_dh_01 = denavit(0, -90, 320, teta1)
```

```
TX40_2_dh_01 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0   320
     0     0     0     1
```

```


% Matriz parcial 1-2:
TX40_2_dh_12 = denavit(225, 0, 35, teta2-90)
```

```
TX40_2_dh_12 = 4×4
     0     1     0     0
    -1     0     0  -225
     0     0     1    35
     0     0     0     1
```

```
% Matriz parcial 2-3:
TX40_2_dh_23 = denavit(0, -90, 0, teta3-90)
```

```
TX40_2_dh_23 = 4×4
    0.7071         0    0.7071         0
   -0.7071         0    0.7071         0
         0   -1.0000         0         0
         0         0         0    1.0000
```

```
% Matriz parcial 3-4:
TX40_2_dh_34 = denavit(0, 90, 225, teta4)
```

```
TX40_2_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   225
     0     0     0     1
```

```
% Matriz parcial 4-5:
TX40_2_dh_45 = denavit(0, -90, 0, teta5)
```

```
TX40_2_dh_45 = 4×4
     0     0     1     0
    -1     0     0     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
TX40_2_dh_56 = denavit(0, 0, 65, teta6+180)
```

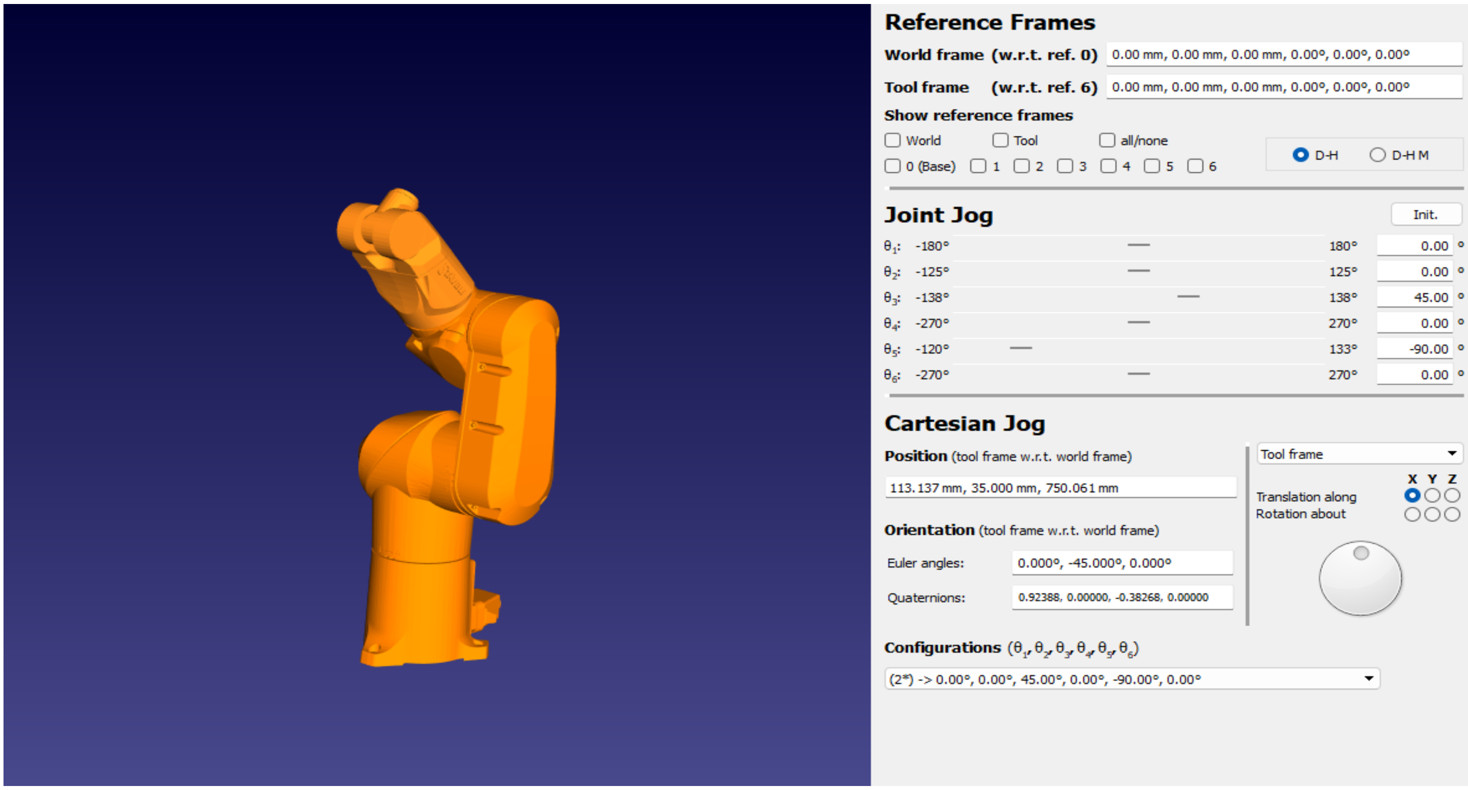
```
TX40_2_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1    65
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante Staubli TX40 (Validação 2):')
```

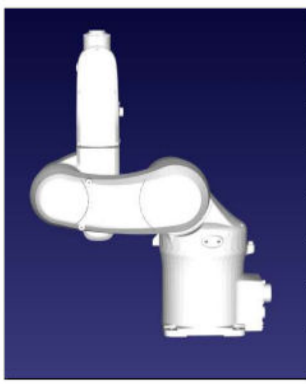
```
Matriz Resultante Staubli TX40 (Validação 2):
```

```
TX40_2_MR = TX40_2_dh_01 * TX40_2_dh_12 * TX40_2_dh_23 * TX40_2_dh_34 *
TX40_2_dh_45 * TX40_2_dh_56
```

```
TX40_2_MR = 4×4
    0.7071         0   -0.7071  113.1371
         0    1.0000         0   35.0000
    0.7071         0    0.7071  750.0610
         0         0         0    1.0000
```

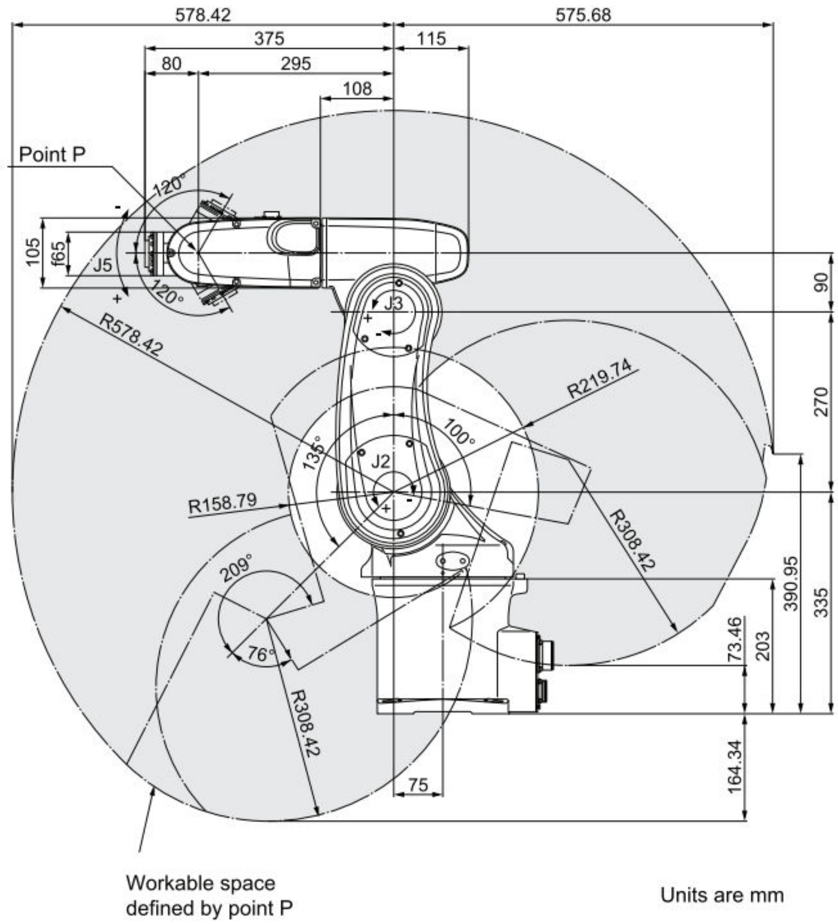


Manipulador Omron Adept Viper s650



Omron Adept Viper s650				
Link	**a**	**alfa**	**d**	**teta**
0 - 1	75	-90	335	teta1
1 - 2	270	0	0	teta2
2 - 3	90	-90	0	teta3 + 180
3 - 4	0	90	295	teta4
4 - 5	0	-90	0	teta5
5 - 6	0	0		teta6 + 180

Dimensions Omron Adept Viper s650 (mm)




```
% Validação 1 (angulos iguais a zero):
teta1 = 0;
teta2 = 0;
teta3 = 0;
teta4 = 0;
teta5 = 0;
teta6 = 0;
fprintf('Omron Adept Viper s650 - Validação 1 \n');
```

```
Omron Adept Viper s650 - Validação 1
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 0 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = 0 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais s650 (Validação 1):')
```

```
Matrizes parciais s650 (Validação 1):
```

```
% Matriz parcial 0-1:
s650_1_dh_01 = denavit(75, -90, 335, teta1)
```

```
s650_1_dh_01 = 4×4
     1     0     0    75
     0     0     1     0
     0    -1     0   335
     0     0     0     1
```

```


% Matriz parcial 1-2:
s650_1_dh_12 = denavit(270, 0, 0, teta2)
```

```
s650_1_dh_12 = 4×4
     1     0     0   270
     0     1     0     0
     0     0     1     0
     0     0     0     1
```

```


% Matriz parcial 2-3:
s650_1_dh_23 = denavit(90, -90, 0, teta3+180)
```

```
s650_1_dh_23 = 4×4
    -1     0     0   -90
     0     0    -1     0
     0    -1     0     0
     0     0     0     1
```

```


% Matriz parcial 3-4:
s650_1_dh_34 = denavit(0, 90, 295, teta4)
```

```
s650_1_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   295
     0     0     0     1
```

```


% Matriz parcial 4-5:
s650_1_dh_45 = denavit(0, -90, 0, teta5)
```

```
s650_1_dh_45 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
s650_1_dh_56 = denavit(0, 0, 80, teta6+180)
```

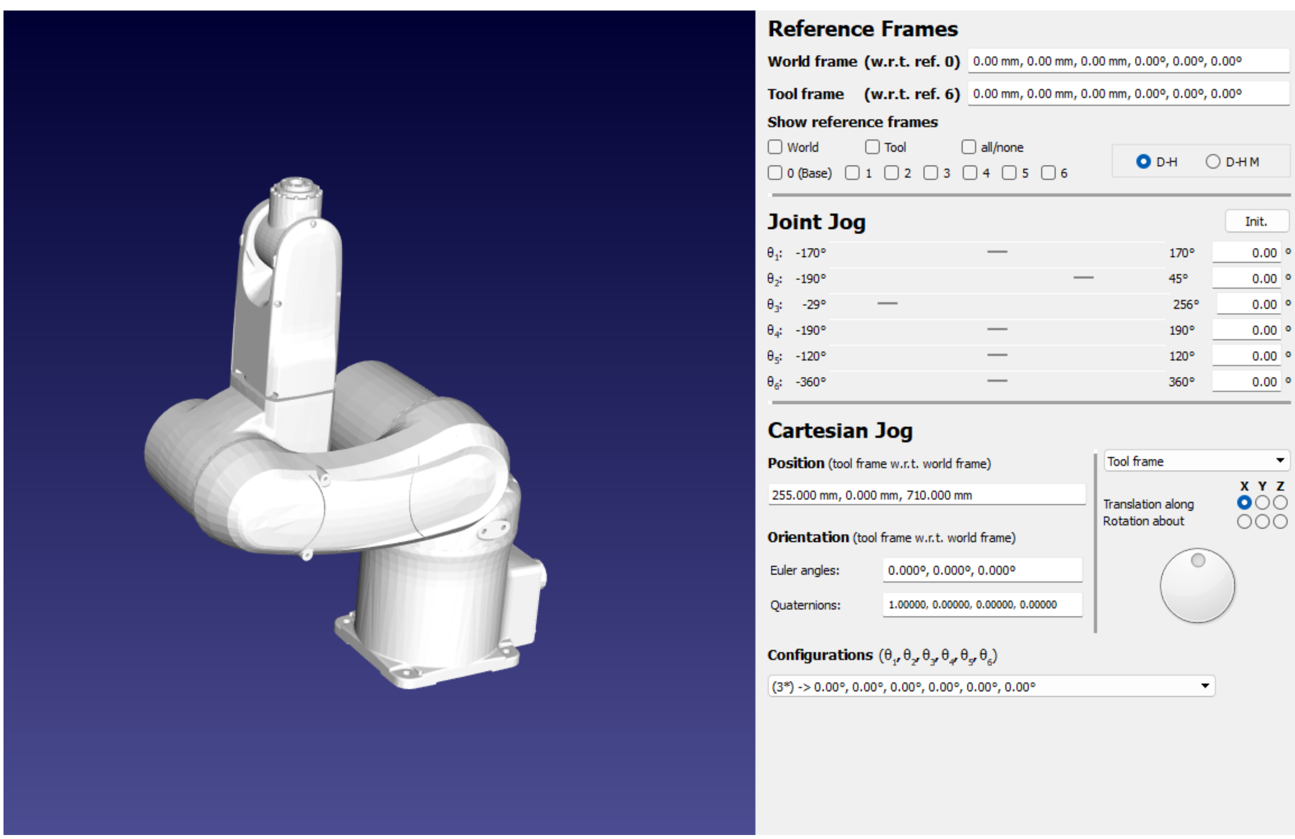
```
s650_1_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1    80
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante Omron Adept Viper s650 (Validação 1):')
```

```
Matriz Resultante Omron Adept Viper s650 (Validação 1):
```

```
s650_1_MR = s650_1_dh_01 * s650_1_dh_12 * s650_1_dh_23 * s650_1_dh_34 *
s650_1_dh_45 * s650_1_dh_56
```

```
s650_1_MR = 4×4
     1     0     0   255
     0     1     0     0
     0     0     1   710
     0     0     0     1
```



```
% Validação 2 (angulos teta3 = -20 e teta5 = 80):
teta1 = 0;
teta2 = 0;
teta3 = -20;
teta4 = 0;
teta5 = 80;
teta6 = 0;
fprintf('Omron Adept Viper s650 - Validação 2 \n');
```

```
Omron Adept Viper s650 - Validação 2
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = -20 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = 80 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais s650 (Validação 2):')
```

```
Matrizes parciais s650 (Validação 2):
```

```
% Matriz parcial 0-1:
s650_2_dh_01 = denavit(75, -90, 335, teta1)
```

```
s650_2_dh_01 = 4×4
     1     0     0    75
     0     0     1     0
     0    -1     0   335
     0     0     0     1
```

```


% Matriz parcial 1-2:
s650_2_dh_12 = denavit(270, 0, 0, teta2)
```

```
s650_2_dh_12 = 4×4
     1     0     0   270
     0     1     0     0
```

```
     0     0     1     0
     0     0     0     1
```

```
% Matriz parcial 2-3:
s650_2_dh_23 = denavit(90, -90, 0, teta3+180)
```

```
s650_2_dh_23 = 4×4
   -0.9397         0   -0.3420  -84.5723
    0.3420         0   -0.9397   30.7818
         0   -1.0000         0         0
         0         0         0    1.0000
```

```
% Matriz parcial 3-4:
s650_2_dh_34 = denavit(0, 90, 295, teta4)
```

```
s650_2_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   295
     0     0     0     1
```

```
% Matriz parcial 4-5:
s650_2_dh_45 = denavit(0, -90, 0, teta5)
```

```
s650_2_dh_45 = 4×4
    0.1736         0   -0.9848         0
    0.9848         0    0.1736         0
         0   -1.0000         0         0
         0         0         0    1.0000
```

```
% Matriz parcial 5-6:
s650_2_dh_56 = denavit(0, 0, 80, teta6+180)
```

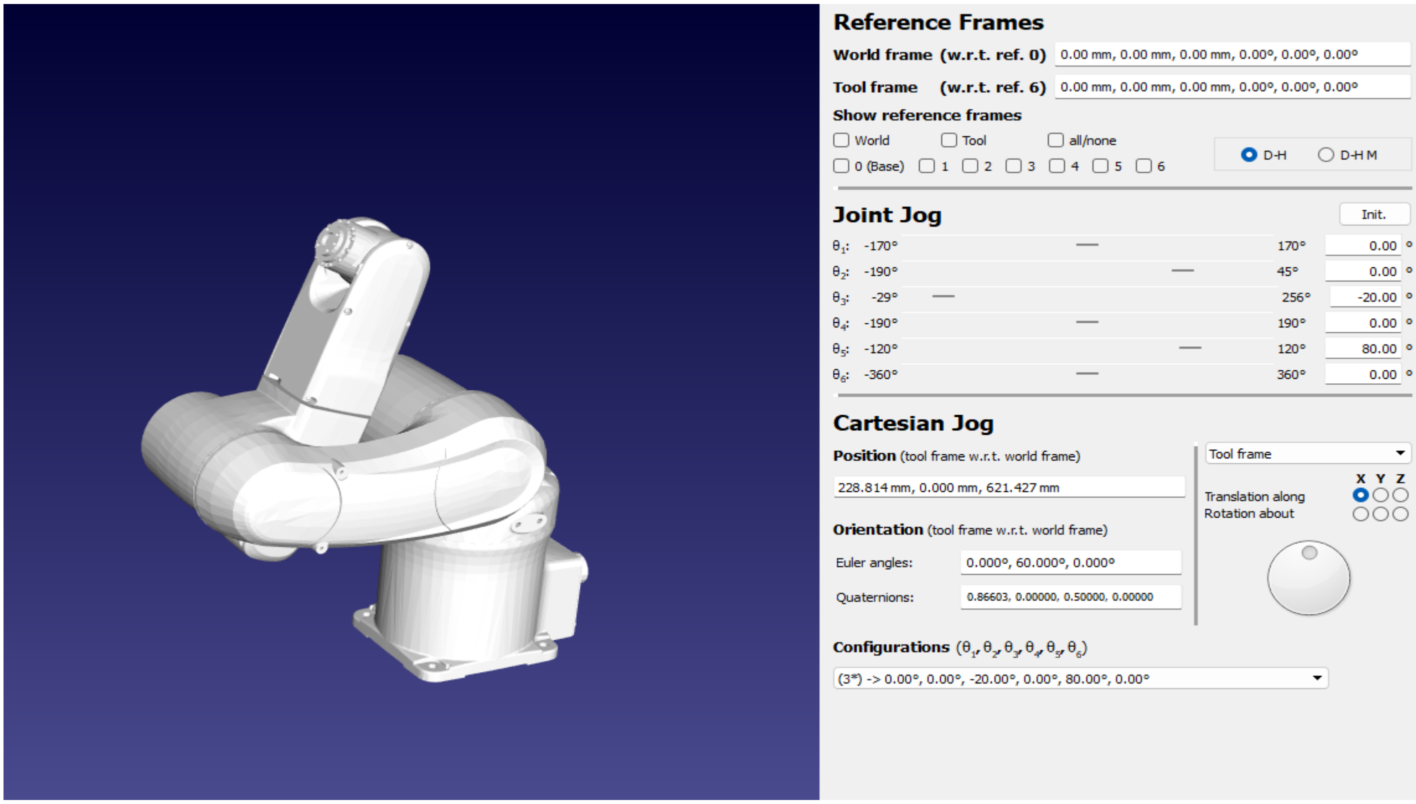
```
s650_2_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1    80
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante Omron Adept Viper s650 (Validação 2):')
```

```
Matriz Resultante Omron Adept Viper s650 (Validação 2):
```

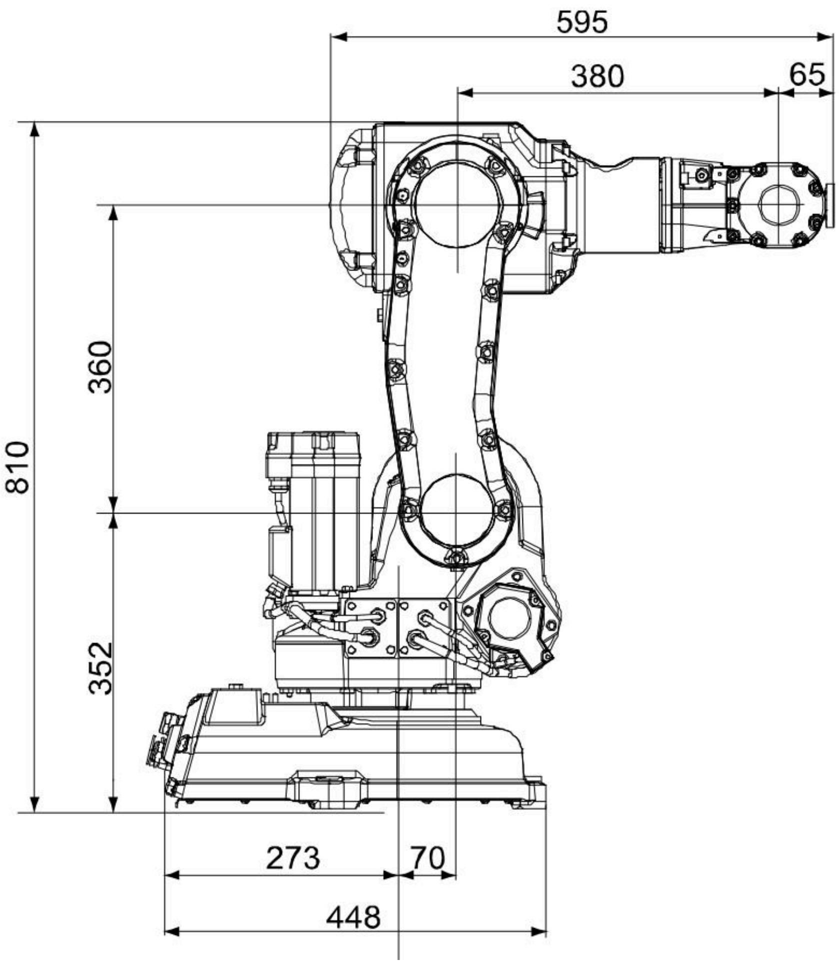
```
s650_2_MR = s650_2_dh_01 * s650_2_dh_12 * s650_2_dh_23 * s650_2_dh_34 *
s650_2_dh_45 * s650_2_dh_56
```

```
s650_2_MR = 4×4
    0.5000         0    0.8660  228.8138
         0    1.0000         0         0
   -0.8660         0    0.5000  621.4275
         0         0         0    1.0000
```



Manipulador ABB IRB140

ABB IRB140				
Link	**a**	**alfa**	**d**	**teta**
0 - 1	70	-90	352	teta1
1 - 2	360	0	0	teta2 - 90
2 - 3	0	-90	0	teta3
3 - 4	0	90	380	teta4
4 - 5	0	-90	0	teta5
5 - 6	0	0	65	teta6 + 180




```
% Validação 1 (angulos iguais a zero):
teta1 = 0;
teta2 = 0;
teta3 = 0;
teta4 = 0;
teta5 = 0;
teta6 = 0;
fprintf('ABB IRB140 - Validação 1 \n');
```

```
ABB IRB140 - Validação 1
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 0 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = 0 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais IRB140 (Validação 1):')
```

```
Matrizes parciais IRB140 (Validação 1):
```

```
% Matriz parcial 0-1:
IRB140_1_dh_01 = denavit(70, -90, 352, teta1)
```

```
IRB140_1_dh_01 = 4×4
     1     0     0    70
     0     0     1     0
     0    -1     0   352
     0     0     0     1
```

```
% Matriz parcial 1-2:
IRB140_1_dh_12 = denavit(360, 0, 0, teta2-90)
```

```
IRB140_1_dh_12 = 4×4
     0     1     0     0
    -1     0     0  -360
     0     0     1     0
     0     0     0     1
```

```
% Matriz parcial 2-3:
IRB140_1_dh_23 = denavit(0, -90, 0, teta3)
```

```
IRB140_1_dh_23 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 3-4:
IRB140_1_dh_34 = denavit(0, 90, 380, teta4)
```

```
IRB140_1_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   380
     0     0     0     1
```

```
% Matriz parcial 4-5:
IRB140_1_dh_45 = denavit(0, -90, 0, teta5)
```

```
IRB140_1_dh_45 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
IRB140_1_dh_56 = denavit(0, 0, 65, teta6+180)
```

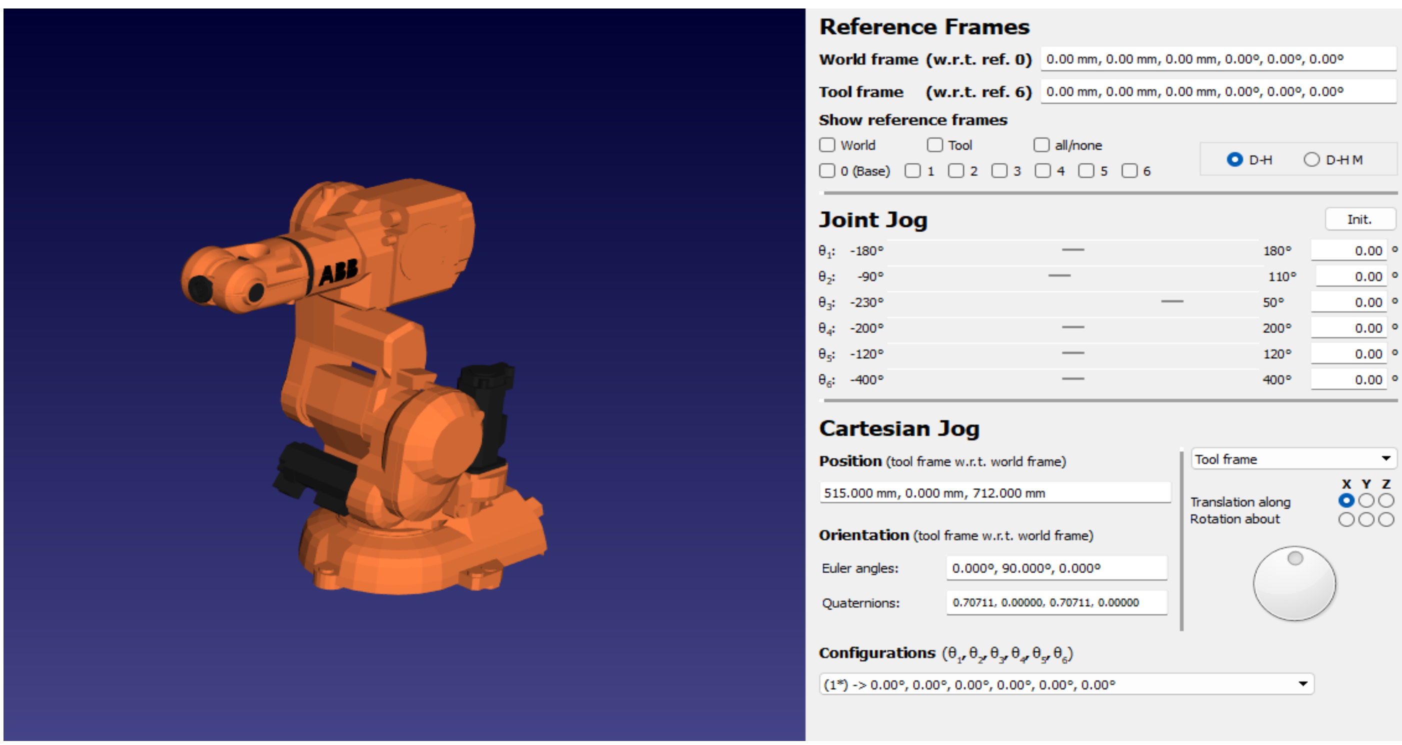
```
IRB140_1_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1    65
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante ABB IRB140 (Validação 1):')
```

```
Matriz Resultante ABB IRB140 (Validação 1):
```

```
IRB140_1_MR = IRB140_1_dh_01 * IRB140_1_dh_12 * IRB140_1_dh_23 * IRB140_1_dh_34 *
IRB140_1_dh_45 * IRB140_1_dh_56
```

```
IRB140_1_MR = 4×4
     0     0     1   515
     0     1     0     0
    -1     0     0   712
     0     0     0     1
```




```
% Validação 2 (angulos teta3 = 45 e teta5 = -90):
teta1 = 0;
teta2 = 0;
```

```
teta3 = 45;
teta4 = 0;
teta5 = -90;
teta6 = 0;
fprintf('ABB IRB140 - Validação 2 \n');
```

```
ABB IRB140 - Validação 2
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 45 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = -90 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais IRB140 (Validação 2):')
```

```
Matrizes parciais IRB140 (Validação 2):
```

```
% Matriz parcial 0-1:
IRB140_2_dh_01 = denavit(70, -90, 352, teta1)
```

```
IRB140_2_dh_01 = 4×4
     1     0     0    70
     0     0     1     0
     0    -1     0   352
     0     0     0     1
```

```


% Matriz parcial 1-2:
IRB140_2_dh_12 = denavit(360, 0, 0, teta2-90)
```

```
IRB140_2_dh_12 = 4×4
     0     1     0     0
    -1     0     0  -360
     0     0     1     0
     0     0     0     1
```

```
% Matriz parcial 2-3:
IRB140_2_dh_23 = denavit(0, -90, 0, teta3)
```

```
IRB140_2_dh_23 = 4×4
    0.7071         0   -0.7071         0
    0.7071         0    0.7071         0
         0   -1.0000         0         0
         0         0         0    1.0000
```

```
% Matriz parcial 3-4:
IRB140_2_dh_34 = denavit(0, 90, 380, teta4)
```

```
IRB140_2_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   380
     0     0     0     1
```

```
% Matriz parcial 4-5:
IRB140_2_dh_45 = denavit(0, -90, 0, teta5)
```

```
IRB140_2_dh_45 = 4×4
     0     0     1     0
    -1     0     0     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
IRB140_2_dh_56 = denavit(0, 0, 65, teta6+180)
```

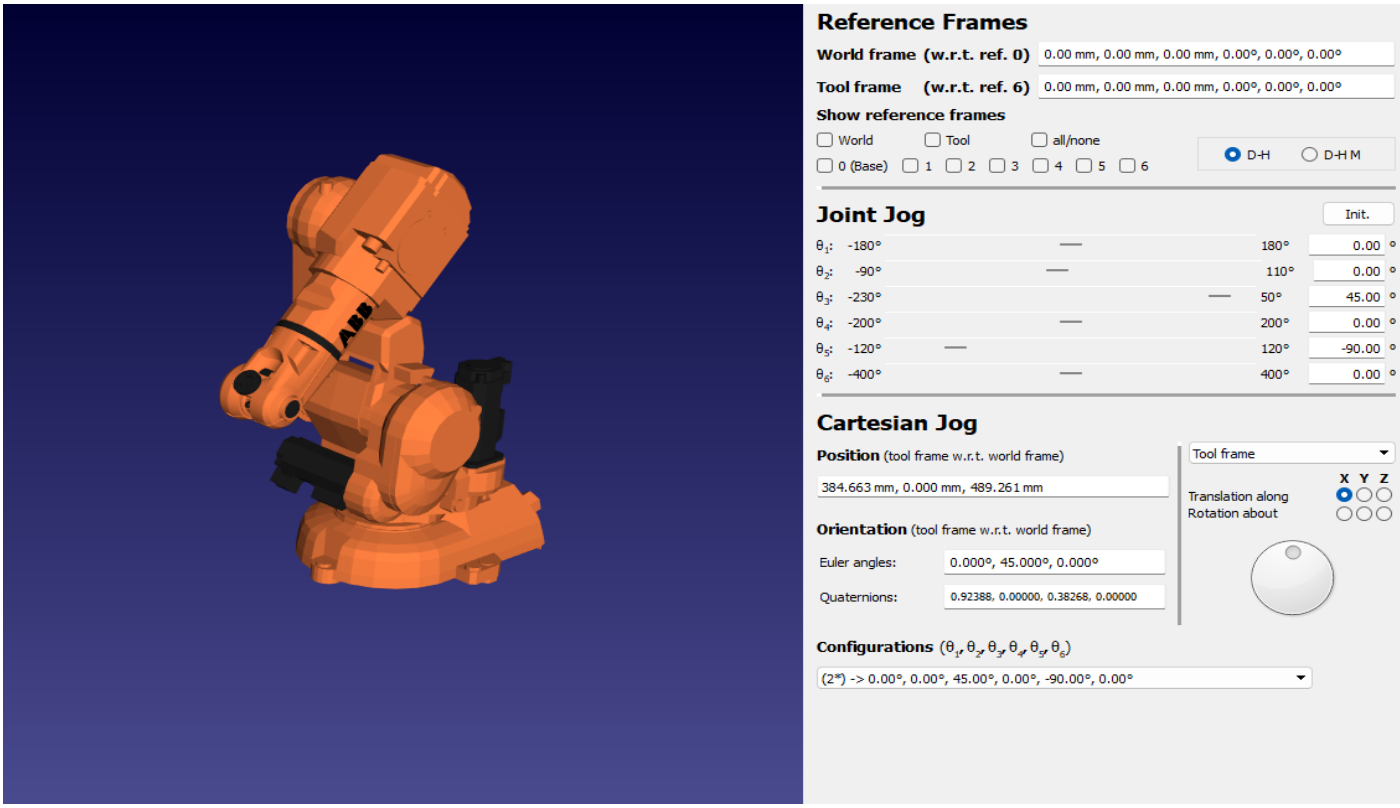
```
IRB140_2_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1    65
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante ABB IRB140 (Validação 2):')
```

```
Matriz Resultante ABB IRB140 (Validação 2):
```

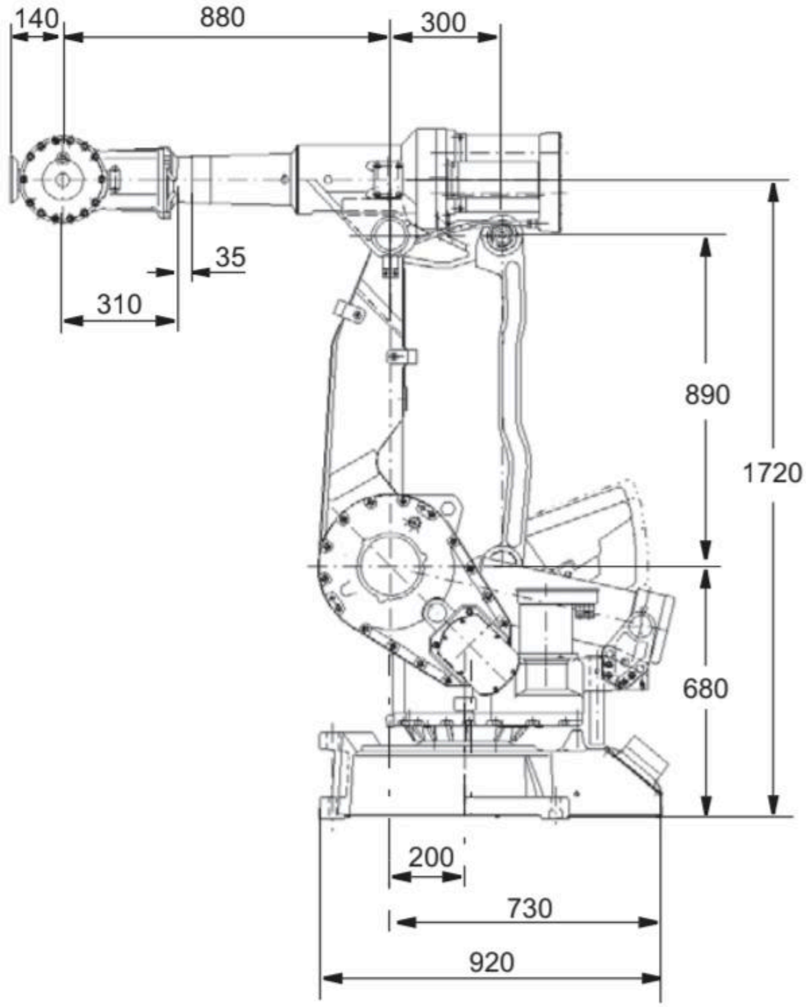
```
IRB140_2_MR = IRB140_2_dh_01 * IRB140_2_dh_12 * IRB140_2_dh_23 * IRB140_2_dh_34 *
IRB140_2_dh_45 * IRB140_2_dh_56
```

```
IRB140_2_MR = 4×4
    0.7071         0    0.7071  384.6625
         0    1.0000         0         0
   -0.7071         0    0.7071  489.2614
         0         0         0    1.0000
```



Manipulador ABB IRB4400

ABB IRB4400				
Link	**a**	**alfa**	**d**	**teta**
0 - 1	200	-90	680	teta1
1 - 2	890	0	0	teta2 - 90
2 - 3	150	-90	0	teta3
3 - 4	0	90	880	teta4
4 - 5	0	-90	0	teta5
5 - 6	0	0	140	teta6 + 180




```
% Validação 1 (angulos iguais a zero):
teta1 = 0;
teta2 = 0;
teta3 = 0;
teta4 = 0;
teta5 = 0;
teta6 = 0;
fprintf('ABB IRB4400 - Validação 1 \n');
```

```
ABB IRB4400 - Validação 1
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 0 ;
```

```matlab
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```matlab
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = 0 ;
```

```matlab
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```matlab
fprintf('\n\n');

disp('Matrizes parciais IRB4400 (Validação 1):')
```

```
Matrizes parciais IRB4400 (Validação 1):
```

```matlab
% Matriz parcial 0-1:
IRB4400_1_dh_01 = denavit(200, -90, 680, teta1)
```

```
IRB4400_1_dh_01 = 4×4
     1     0     0   200
     0     0     1     0
     0    -1     0   680
     0     0     0     1
```

```matlab

% Matriz parcial 1-2:
IRB4400_1_dh_12 = denavit(890, 0, 0, teta2-90)
```

```
IRB4400_1_dh_12 = 4×4
     0     1     0     0
    -1     0     0  -890
     0     0     1     0
     0     0     0     1
```

```matlab

% Matriz parcial 2-3:
IRB4400_1_dh_23 = denavit(150, -90, 0, teta3)
```

```
IRB4400_1_dh_23 = 4×4
     1     0     0   150
     0     0     1     0
     0    -1     0     0
     0     0     0     1
```

```matlab

% Matriz parcial 3-4:
IRB4400_1_dh_34 = denavit(0, 90, 880, teta4)
```

```
IRB4400_1_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   880
     0     0     0     1
```

```
% Matriz parcial 4-5:
IRB4400_1_dh_45 = denavit(0, -90, 0, teta5)
```

```
IRB4400_1_dh_45 = 4×4
     1     0     0     0
     0     0     1     0
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
IRB4400_1_dh_56 = denavit(0, 0, 140, teta6+180)
```

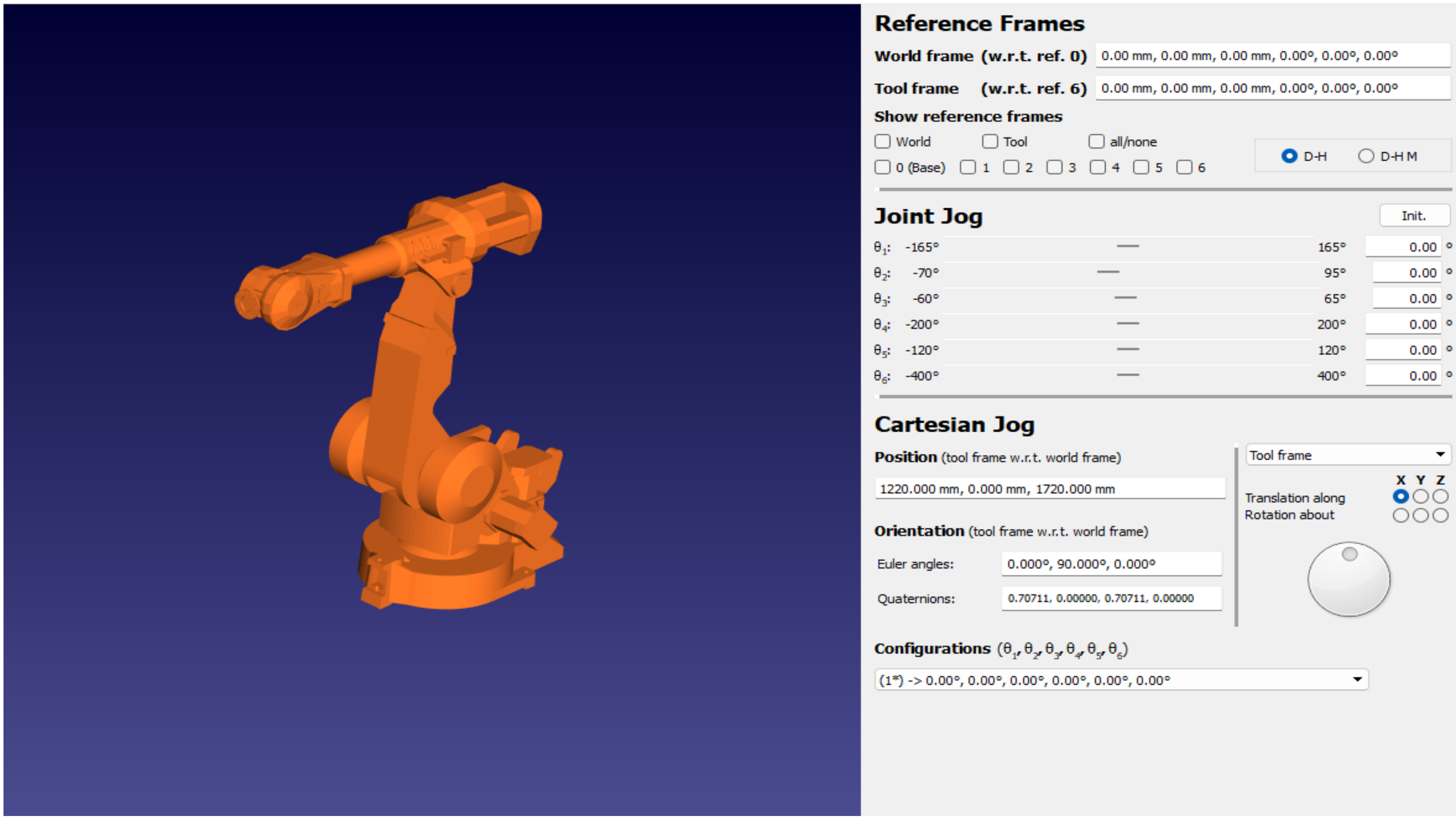
```
IRB4400_1_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1   140
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante ABB IRB4400 (Validação 1):')
```

```
Matriz Resultante ABB IRB4400 (Validação 1):
```

```
IRB4400_1_MR = IRB4400_1_dh_01 * IRB4400_1_dh_12 * IRB4400_1_dh_23 *
IRB4400_1_dh_34 * IRB4400_1_dh_45 * IRB4400_1_dh_56
```

```
IRB4400_1_MR = 4×4
           0           0           1        1220
           0           1           0           0
          -1           0           0        1720
           0           0           0           1
```




```
% Validação 2 (angulos teta3 = 45 e teta5 = -90):
teta1 = 0;
teta2 = 0;
teta3 = 45;
teta4 = 0;
teta5 = -90;
teta6 = 0;
fprintf('ABB IRB4400 - Validação 2 \n');
```

```
ABB IRB4400 - Validação 2
```

```
fprintf('teta1 = %0.f ; ', teta1);
```

```
teta1 = 0 ;
```

```
fprintf('teta2 = %0.f ; ', teta2);
```

```
teta2 = 0 ;
```

```
fprintf('teta3 = %0.f ; ', teta3);
```

```
teta3 = 45 ;
```

```
fprintf('teta4 = %0.f ; ', teta4);
```

```
teta4 = 0 ;
```

```
fprintf('teta5 = %0.f ; ', teta5);
```

```
teta5 = -90 ;
```

```
fprintf('teta6 = %0.f ', teta6);
```

```
teta6 = 0
```

```
fprintf('\n\n');

disp('Matrizes parciais IRB4400 (Validação 2):')
```

```
Matrizes parciais IRB4400 (Validação 2):
```

```
% Matriz parcial 0-1:
IRB4400_2_dh_01 = denavit(200, -90, 680, teta1)
```

```
IRB4400_2_dh_01 = 4×4
     1     0     0   200
     0     0     1     0
     0    -1     0   680
     0     0     0     1
```

```
% Matriz parcial 1-2:
IRB4400_2_dh_12 = denavit(890, 0, 0, teta2-90)
```

```
IRB4400_2_dh_12 = 4×4
     0     1     0     0
    -1     0     0  -890
     0     0     1     0
     0     0     0     1
```

```
% Matriz parcial 2-3:
IRB4400_2_dh_23 = denavit(150, -90, 0, teta3)
```

```
IRB4400_2_dh_23 = 4×4
    0.7071         0   -0.7071  106.0660
    0.7071         0    0.7071  106.0660
         0   -1.0000         0         0
         0         0         0    1.0000
```

```
% Matriz parcial 3-4:
IRB4400_2_dh_34 = denavit(0, 90, 880, teta4)
```

```
IRB4400_2_dh_34 = 4×4
     1     0     0     0
     0     0    -1     0
     0     1     0   880
     0     0     0     1
```

```
% Matriz parcial 4-5:
IRB4400_2_dh_45 = denavit(0, -90, 0, teta5)
```

```
IRB4400_2_dh_45 = 4×4
     0     0     1     0
    -1     0     0     0
```

```
     0    -1     0     0
     0     0     0     1
```

```
% Matriz parcial 5-6:
IRB4400_2_dh_56 = denavit(0, 0, 140, teta6+180)
```

```
IRB4400_2_dh_56 = 4×4
    -1     0     0     0
     0    -1     0     0
     0     0     1   140
     0     0     0     1
```

```
% Matriz Final:
disp('Matriz Resultante ABB IRB4400 (Validação 2):')
```

```
Matriz Resultante ABB IRB4400 (Validação 2):
```

```
IRB4400_2_MR = IRB4400_2_dh_01 * IRB4400_2_dh_12 * IRB4400_2_dh_23 *
IRB4400_2_dh_34 * IRB4400_2_dh_45 * IRB4400_2_dh_56
```

```
IRB4400_2_MR = 4×4
1.0e+03 *
    0.0007         0    0.0007    1.0273
         0    0.0010         0         0
   -0.0007         0    0.0007    1.1528
         0         0         0    0.0010
```